

Functional Approximation Methods for Differentially Private Distribution Estimation

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Abstract—The cumulative distribution function (CDF) is fundamental for characterizing random variables, making it essential in applications that require privacy-preserving data analysis. This paper introduces a novel framework for constructing differentially private CDFs inspired by functional analysis and the functional mechanism. We develop two variants: a polynomial projection method, which projects the empirical CDF into a polynomial space, and a sparse approximation method via matching pursuit, which projects it into arbitrary function spaces constructed from dictionaries. In both cases, the empirical CDF is approximated within the chosen space, and the corresponding coefficients are privatized to guarantee differential privacy. Compared with existing approaches such as histogram queries, tree-based methods, and adaptive quantiles, our approach achieves comparable or superior performance. Our methods are particularly well-suited to decentralized settings and scenarios where CDFs must be efficiently updated with newly collected or streaming data. They are also readily extensible to multivariate distributions while remaining effective. In addition, we investigate the influence of parameters such as dictionary size and systematically evaluate different dictionary constructions, including Legendre polynomials, B-splines, and distribution-based functions. Overall, our contributions advance the development of practical and reliable methods for privacy-preserving CDF estimation.

Index Terms—Differential privacy, functional analysis, cumulative distribution function approximation, dictionary learning.

I. INTRODUCTION

THE cumulative distribution function (CDF) is a fundamental object in statistical analysis, serving as a cornerstone in both classical statistics and modern machine learning. For instance, in hypothesis testing, test statistics are compared against critical values derived from the CDF to guide decisions about hypotheses. Furthermore, CDFs play a central role in risk assessment and decision-making under uncertainty, as they provide a comprehensive characterization of the distribution of possible outcomes. When a distribution is not known and only sampled observed data are available, we typically use the empirical cumulative distribution function (eCDF) to estimate the unknown true CDF.

In this paper we provide new methods for approximating a CDF when the data is sensitive or private. More specifically,

we design new approaches for estimating CDFs under differential privacy (DP) constraints [2]. This paper is motivated by two related application scenarios. In federated learning, the sensitive data is spread across many data holders. For example, different groups may have subject data from human health research, and summary statistics such as a CDF can reveal aggregate features of the population distribution. An aggregator collects a DP estimate from each group's local data to form this population summary. The second is when data comes in over time in batches. A DP CDF estimate is updated and published as more data is collected. Central to both of these tasks is DP CDF estimation at a single site: ideally we want an estimation method which can be aggregated or updated over time.

The simplest method for approximating a CDF under DP is to histogram the data using a fixed binning [3]. This effectively quantizes the data: a histogram is a collection of count queries for each bin. Adding noise to the bin counts guarantees DP, and we can compute an eCDF using the quantized values. However, fixed binning may lead to suboptimal approximation accuracy due to the discretization of the data domain. To alleviate this limitation, adaptive histogram boundaries [4] dynamically adjust bin widths based on data density. This approach assumes that the data are pre-quantized or that the data domain is a subset of the integers. Hierarchical histograms [5], [6] improve the resolution of the binning by building a tree structure on a fine partitioning of the data, thereby mitigating the limitations of fixed binning. Additional discretization-based methods have been studied under various settings. For instance, in large-scale federated settings with a massive number of sites or users, random subsampling has been used to construct histograms efficiently [7]. Other work has considered multidimensional data using transforms such as wavelets [8], [9]. Related distribution estimation problems have also been investigated under alternative privacy models, including local and shuffle models [10]–[12]. Beyond histogram-based approaches, adaptive quantile mechanisms [13] provide an alternative approach to CDF estimation. Instead of relying on histogram counts, these methods estimate private quantiles. However, these adaptive methods typically incur high communication costs due to their iterative nature, which limits their applicability in decentralized settings. While differentially private estimation of probability density functions (PDFs) has also been studied [14]–[16], we focus on the CDF, as it allows direct access to ranks, quantiles, thresholds, and other statistics: estimating these from DP PDFs may involve integrating the private PDF, making it less straightforward to provide approximation guarantees.

Given the vast work on DP statistical estimation, why do we

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need new methods for DP CDF estimation? Existing methods lack flexibility or efficiency in certain scenarios, such as decentralized settings or streaming data updates. For example, in decentralized settings, approaches like adaptive quantiles involve multiple communication rounds, while a more efficient alternative is to allow each site to transmit its information once for centralized aggregation into a global DP CDF. In streaming scenarios, methods such as adaptive quantiles require accessing old data when incorporating new samples, leading to repeated noise addition and increased privacy loss. Histogram-based methods may require recomputing the bin counts or rebuilding the histogram structure when new data arrive, making them inefficient for continuous updates. Motivated by these limitations, in this work we propose a novel framework for constructing differentially private CDFs in which the empirical CDF is projected into an appropriate function space and approximated in a manner analogous to standard signal decomposition techniques [17]–[19]. The coefficients of these functions are then privatized to ensure differential privacy. Within this framework, we introduce two approaches. The first approach, called the polynomial projection method, projects the eCDF into a polynomial space using families of orthogonal polynomials commonly employed in functional analysis [20]–[23], followed by application of the functional mechanism to ensure differential privacy [24]. The second approach, a sparse approximation method via matching pursuit, constructs arbitrary function spaces from dictionaries instead of relying on polynomials, enabling flexible approximation of more complex CDF shapes. Our primary focus is one-dimensional CDF estimation, although the proposed framework naturally extends to high-dimensional CDF estimation, as demonstrated in our experiments.

The contributions of this work are:

- We develop two methods for approximating empirical CDFs via projections into function spaces: a polynomial projection method based on a fixed polynomial function space, and a sparse approximation method via matching pursuit that generalizes this idea by allowing flexible function spaces, together forming a general framework for privacy-preserving CDF estimation and offering a novel perspective on the problem.
- We provide theoretical analysis for approximation using orthonormal families, including upper bounds on the estimation error between the DP CDFs and true CDFs, and investigate the role of post-processing, showing that it preserves the validity of the CDF without compromising the approximation performance.
- We demonstrate that our methods achieve comparable or superior performance to existing techniques across a range of scenarios. In particular, our approaches are well-suited for decentralized settings, and when updating the CDF with newly collected data, they outperform alternatives by avoiding the need to access previously collected data, thereby conserving the privacy budget. We further evaluate the applicability of our framework to high-dimensional CDF estimation, demonstrating the effectiveness of the sparse approximation method in such

settings.

- We explore the effects of key parameters, such as dictionary size, and systematically examine different dictionary constructions, including polynomial, B-spline, and parametric distribution-based bases for function spaces, demonstrating the flexibility and robustness of our methods.

II. BACKGROUND

A. Differential Privacy

Differential privacy has been developed to enable statistical analyses on datasets while preserving the privacy of individuals included in the data. It is defined in terms of neighboring databases; two sets are considered neighboring if they differ by a single entry.

Definition 1 ((ϵ, δ) -DP [25]). *Let $\epsilon, \delta \geq 0$, a randomized algorithm $M : \mathcal{D} \rightarrow \mathcal{Y}$ is (ϵ, δ) -differentially private if for all $Y \subseteq \mathcal{Y}$ and for all neighboring datasets $D, D' \in \mathcal{D}$,*

$$\mathbb{P}[M(D) \in Y] \leq e^\epsilon \mathbb{P}[M(D') \in Y] + \delta.$$

The parameter ϵ represents the privacy budget, where a smaller value offers stronger privacy but may result in less accurate responses. The parameter δ indicates the probability of information leakage. To design a (ϵ, δ) -DP algorithm that approximates a desired function $f : \mathcal{D} \rightarrow \mathbb{R}^d$, a common approach is to add random noise of appropriate magnitude to the output of $f(D)$ [26]–[33]. For example, when $\epsilon, \delta \in (0, 1)$, the Gaussian mechanism is defined as $M(D) = f(D) + \mathcal{N}(0, \sigma^2 \mathbf{I})$, where $\sigma = \Delta \sqrt{2 \log(1.25/\delta)}/\epsilon$ and $\Delta = \max_{D \sim D'} \|f(D) - f(D')\|_2$ represents ℓ_2 sensitivity of function f .

Differential privacy has several properties that make it particularly useful in applications. The composition property [25], [34]–[38] allows for a modular design of mechanisms: if each component of a mechanism is differentially private, then their composition also preserves differential privacy. The post-processing property ensures the safety of conducting arbitrary computations on the output of a differentially private mechanism: because of a data processing inequality [39], post-processing can only reduce the privacy risk provided by the mechanism.

B. CDF Estimation

In this work, we focus on the problem of estimating a cumulative distribution function from observations of a random variable. Accurate CDF estimation under privacy constraints is crucial in many applications, and various techniques have been developed to achieve this goal while preserving differential privacy. Existing approaches can be broadly categorized into histogram-based methods, hierarchical tree-based methods, and quantile-based methods. These methods typically introduce noise into data-dependent representations, such as histogram counts, hierarchical structures, or estimated quantiles, to achieve differential privacy. In our evaluation, we compare our methods with three representative baselines from these categories:

1) *Histogram Queries*: The histogram queries (HQ) method partitions the data into a fixed number of uniformly spaced bins. The number of observations in each bin is counted, and noise is added to protect the privacy of these counts. The resulting noisy histogram can be used to estimate the CDF in a differentially private manner. To enhance accuracy, post-processing is applied by setting any negative noisy counts to 0, ensuring the estimated CDF is non-decreasing and non-negative.

2) *Tree-Based Histogram*: The tree-based (TB) histogram method extends standard flat histograms by organizing the data domain into a hierarchical structure, where each node stores the count of observations within a specific range. Noisy counts are added to the tree nodes to guarantee differential privacy, and CDF values are estimated by aggregating noisy counts from selected nodes in the hierarchy. A common construction is based on a balanced binary tree over the data domain. Recently, Rameshwar et al. [40] advanced this paradigm by introducing optimized tree-based mechanisms and identifying near-optimal tree heights and branching factors. In our framework, we use this tree-based mechanism as a benchmarking baseline.

3) *Adaptive Quantiles*: The adaptive quantiles (AQ) method provides a more refined approach by leveraging known quantiles to guide the estimation process. Initially, a set of known quantiles $Q = \{0 : a, 1 : b\}$ is established, where a and b represent the lower and upper bounds of the data range. The method proceeds by selecting an initial value $x_0 = (a+b)/2$, counting the number of observations above and below x_0 , and adding noise to these counts to preserve privacy. The corresponding quantile q is then computed and stored as $Q[q] = x_0$. This process is iteratively repeated, with each new candidate x_i selected as the midpoint of the largest interval between known quantiles until the privacy budget is exhausted. Due to the noise added during the process, situations may arise where $x_1 > x_2$ but $q_1 < q_2$. Post-processing is necessary, and one approach is to reorder the x and q values.

TABLE I: Key Notations

Symbol	Definition
F	true CDF of the distribution
F_n	eCDF of samples $\{x_k\}_{k=1}^n$
$\hat{F}$	optimal projection of F onto predefined space
$\hat{F}_n$	optimal projection of F_n onto predefined space
$\tilde{F}_n$	privacy-preserving approximation of F_n
$\hat{F}_n^s$	optimal projection of F_n with sparsity level s
$\tilde{F}_n^s$	privacy-preserving approximation of $\hat{F}_n^s$

III. CDF APPROXIMATION AS FUNCTION APPROXIMATION

Our approach first approximates the eCDF using a weighted combination of predefined functions and subsequently applies a differential privacy technique to protect the coefficients. With the perturbed coefficients and the associated predefined functions, we are able to reconstruct a privacy-preserving eCDF (see Figure 1). To formalize this, we begin by defining the relevant quantities. Key notations are summarized in Table I. Throughout this work, we assume that the data range is known and approximate the empirical CDF in the

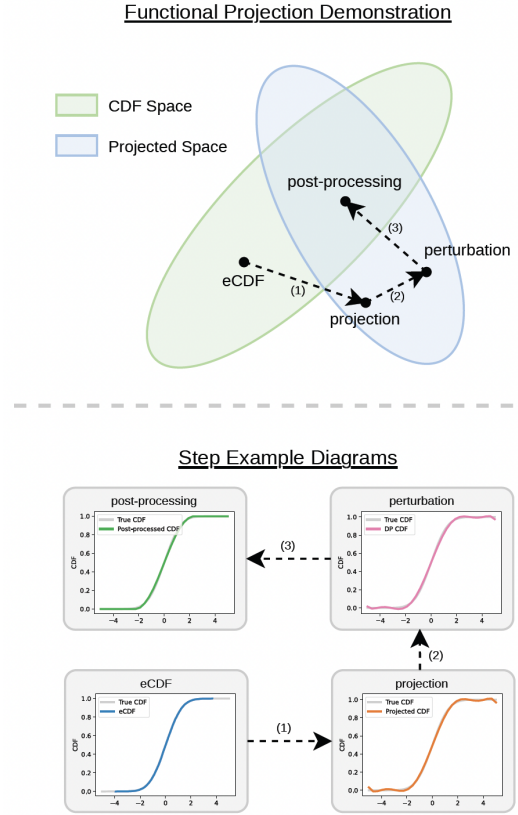


Fig. 1: Illustration of our proposed method. The top panel shows the overview: the eCDF is projected onto the predefined function space, perturbed for differential privacy, and post-processed to obtain a valid DP CDF. The bottom panel presents step-by-step example results for each stage.

Hilbert space $L^2([-A, A])$. When the data range is unknown, existing differentially private techniques can be applied prior to CDF construction, such as privately estimating the domain bounds or truncating the data under approximate differential privacy [41]–[43].

Let F be the CDF of a distribution supported on the interval $[-A, A]$, and let $\{x_k\}_{k=1}^n$ be i.i.d. samples drawn from F , ordered such that $x_1 \leq x_2 \leq \dots \leq x_n$. The eCDF is defined as

$$F_n(x) = \frac{1}{n} \sum_{k=1}^n \mathbb{I}(x_k \leq x),$$

where $\mathbb{I}(x_k \leq x)$ is the indicator function, equal to 1 if $x_k \leq x$, and 0 otherwise. Let $\{\phi_i\}_{i=1}^m$ be an orthonormal basis for an m -dimensional subspace of $L^2([-A, A])$. Define the function space:

$$\mathcal{F} = \text{span}\{\phi_1, \phi_2, \dots, \phi_m\}.$$

The orthogonal projection of F_n onto $\mathcal{F}$ is

$$\hat{F}_n = \arg \min_{f \in \mathcal{F}} \|f - F_n\|_2,$$

which admits the representation

$$\hat{F}_n(x) = \sum_{i=1}^m c_i \phi_i(x).$$

The standard approach to guaranteeing (ϵ, δ) -DP is to perturb the coefficients with noise:

$$\tilde{c}_i = c_i + z_i,$$

where $\{z_i\}_{i=1}^m$ are i.i.d. random variables drawn from an (ϵ, δ) -DP mechanism such as Gaussian or Laplace mechanism. The privacy-preserving approximation of the eCDF is then given by:

$$\tilde{F}_n(x) = \sum_{i=1}^m \tilde{c}_i \phi_i(x).$$

Assuming that $\hat{F}$ is the optimal approximation of F in $\mathcal{F}$ satisfying $\|F - \hat{F}\|_2 \leq \alpha$. The total error of the privacy-preserving estimator $\tilde{F}_n$ can be decomposed as:

$$\begin{aligned} \|F - \tilde{F}_n\|_2 &\leq \|F - F_n\|_2 + \|F_n - \hat{F}_n\|_2 + \|\hat{F}_n - \tilde{F}_n\|_2 \\ &\leq \|F - F_n\|_2 + \|F_n - \hat{F}\|_2 + \|\hat{F}_n - \tilde{F}_n\|_2 \\ &\leq \underbrace{\|F - \hat{F}\|_2}_{\text{approximation error}} + 2 \underbrace{\|F_n - F_n\|_2}_{\text{empirical error}} + \underbrace{\|\hat{F}_n - \tilde{F}_n\|_2}_{\text{privacy error}}. \end{aligned}$$

The first and third inequalities follow from the triangle inequality. The second inequality uses the optimality of $\hat{F}_n$, which ensures that $\|F_n - \hat{F}_n\|_2 \leq \|F_n - \hat{F}\|_2$. This decomposition separates the total error into three components: the approximation error from projecting the true CDF onto the function subspace, the empirical error due to finite sampling, and the privacy error introduced by perturbation. The empirical error can be further bounded as

$$\begin{aligned} \|F - F_n\|_2 &= \left(\int_{-A}^A |F(x) - F_n(x)|^2 dx \right)^{1/2} \\ &\leq \left(\int_{-A}^A \|F - F_n\|_\infty^2 dx \right)^{1/2} \\ &= (2A)^{1/2} \|F - F_n\|_\infty, \end{aligned}$$

where the inequality holds because $|F(x) - F_n(x)| \leq \|F - F_n\|_\infty$ for all $x \in [-A, A]$. The privacy error term can be explicitly bounded in terms of the noise magnitude and the properties of the basis functions:

$$\|\hat{F}_n - \tilde{F}_n\|_2 = \left\| \sum_{i=1}^m z_i \phi_i \right\|_2 \leq \sum_{i=1}^m \|z_i \phi_i\|_2 = \sum_{i=1}^m |z_i| \|\phi_i\|_2,$$

where the inequality follows from triangle inequality. We define the total error threshold τ as a weighted sum of the individual error components:

$$\tau = \alpha + 2\beta + \eta,$$

where α bounds the approximation error, β accounts for the empirical error, and η quantifies the privacy error. The total

error satisfies the following tail bound:

$$\begin{aligned} &\mathbb{P}(\|F - \tilde{F}_n\|_2 > \tau) \\ &\leq \mathbb{P}(\|F - \hat{F}\|_2 + 2\|F - F_n\|_2 + \|\hat{F}_n - \tilde{F}_n\|_2 > \tau) \\ &\leq \mathbb{P}\left(2(2A)^{1/2}\|F - F_n\|_\infty + \sum_{i=1}^m |z_i| \|\phi_i\|_2 > 2\beta + \eta\right) \\ &\leq \mathbb{P}(\|F - F_n\|_\infty > \beta/(2A)^{1/2}) + \mathbb{P}\left(\sum_{i=1}^m |z_i| \|\phi_i\|_2 > \eta\right) \\ &\leq 2 \exp(-2n\beta^2/2A) + \mathbb{P}\left(\sum_{i=1}^m |z_i| > \eta / \max_{1 \leq i \leq m} \|\phi_i\|_2\right). \end{aligned}$$

The last inequality follows from the Dvoretzky-Kiefer-Wolfowitz (DKW) inequality [44], which provides a finite-sample bound on the deviation between the empirical distribution function F_n and the true distribution function F . The bound for privacy error depends on the differential privacy mechanism. For instance, if the Gaussian mechanism is applied, where the noise vector $\mathbf{z} = [z_1, z_2, \dots, z_m]^\top$ is drawn from $\mathcal{N}(0, \sigma^2 \mathbf{I})$, we have $\mathbb{E}[|z_i|] = \sigma\sqrt{2/\pi}$. By the linearity of expectation, we know:

$$\mathbb{E}\left[\sum_{i=1}^m |z_i|\right] = \sum_{i=1}^m \mathbb{E}[|z_i|] = m\sigma\sqrt{2/\pi}.$$

Define $\eta' = \eta / \max_{1 \leq i \leq m} \|\phi_i\|_2$. Applying Markov's inequality yields:

$$\mathbb{P}\left(\sum_{i=1}^m |z_i| \geq \eta'\right) \leq \frac{\mathbb{E}[\sum_{i=1}^m |z_i|]}{\eta'} = \frac{m\sigma\sqrt{2/\pi}}{\eta'}.$$

An even tighter bound can be obtained by applying the Chernoff bound and leveraging the moment generating function of the random variable $\|\mathbf{z}\|_1$ ¹.

For more general L^p approximation spaces (e.g., L^1), the orthogonal projection framework available in the Hilbert space L^2 is generally unavailable. Nevertheless, convergence and approximation error bounds for polynomial approximation in L^p norms can still be established through classical approximation theory in Banach spaces [45]. Extending the proposed framework and its theoretical analysis to more general function spaces remains an interesting direction for future work.

IV. POLYNOMIAL PROJECTION

Building on the theoretical intuition from Section III, we propose a polynomial projection (PP) method to obtain a differentially private approximation of the eCDF. Without loss of generality, we assume $A = 1$. If the sample range lies outside $[-1, 1]$, we first scale the data to fit within this interval. For example, if the data range is $[-A, A]$, each x_k is divided by A . The bounds can be determined either through prior knowledge or by applying differentially private estimation techniques [46], [47], which require additional privacy budget. After obtaining the DP eCDF, the scaling can be reversed to map the result back to the original data range. Section IV-A

¹A detailed derivation can be found at <https://www.ryanhmckenna.com/2021/12/tail-bounds-on-sum-of-half-normal.html>.

presents the polynomial projection approximation in the non-private setting, while Section IV-B introduces its differentially private counterpart.

A. Non-private Polynomial Projection

In this work, we use the Legendre polynomials, although other polynomial families (e.g., Legendre, Chebyshev) can also be employed. The first m functions from any such family span the same subspace of polynomials of degree less than m . Thus, the choice of polynomial family does not affect the projection space itself, but may impact numerical stability and computational efficiency [48]. Appendix A in [49] provides a brief discussion on the selection of polynomial families and other function families, such as Fourier-based approaches.

We leverage the projection theorem [20] (see Theorem 7 in Appendix B) to identify the optimal approximation of the eCDF within the selected polynomial space $\mathcal{P}$. This formulation allows straightforward computation of the projection coefficients, which are related to the moments of the variables. These moment-based coefficients facilitate sensitivity analysis and the calibration of noise for privacy protection.

In the context of the Hilbert space $L^2[-1, 1]$, we establish a polynomial space $\mathcal{P}$ with the basis $\{P_0, P_1, \dots, P_m\}$, where each P_i is a Legendre polynomial of degree i over the interval $[-1, 1]$. This space $\mathcal{P}$ is equipped with an inner product defined as $\langle f_1, f_2 \rangle = \int_{-1}^1 f_1(x)f_2(x) dx$ and a norm given by $\|f\|_2 = \left(\int_{-1}^1 f(x)^2 dx \right)^{1/2}$. The Legendre polynomials are orthogonal to each other. Since every finite-dimensional inner product space is also a Hilbert space, and the Legendre polynomials $\{P_i\}_{i=0}^\infty$ form a complete orthogonal sequence in $L^2[-1, 1]$, $\mathcal{P}$ is a closed subspace of $L^2[-1, 1]$. Given that $\int_{-1}^1 |F_n(x)|^2 dx < \infty$, the eCDF $F_n(x)$ belongs to the $L^2[-1, 1]$ space. According to the projection theorem (Theorem 7), we can identify a unique vector $\hat{F}_n$ in the space $\mathcal{P}$, which is the optimal approximation of the eCDF, as demonstrated in Theorem 1 (see Appendix B for proof).

Theorem 1 (Optimal Approximation of eCDF). *Consider the polynomial space $\mathcal{P}$ spanned by the Legendre polynomials $\{P_0, P_1, \dots, P_m\}$, where each P_i is of degree i . The optimal approximation $\hat{F}_n$ of the eCDF within $\mathcal{P}$ is given by*

$$\hat{F}_n(x) = \sum_{i=0}^m \langle F_n, e_i \rangle e_i = \sum_{i=0}^m \alpha_i \sum_{j=0}^i (\beta_{i,j} (1 - \mu_{j+1})) e_i,$$

where $\alpha_i = 2^i \sqrt{\frac{2i+1}{2}}$, $\beta_{i,j} = \frac{1}{j+1} \binom{i}{j} \binom{i+j-1}{i}$, $\mu_{j+1} = \frac{1}{n} \sum_{k=1}^n x_k^{j+1}$ represents mean of the $(j+1)$ -th moment of the data, and $e_i = \sqrt{\frac{2i+1}{2}} P_i$ is the orthonormal basis of $\mathcal{P}$.

B. Privacy-preserving Polynomial Projection

To achieve a privacy-preserving eCDF, we protect the coefficients $\langle F_n, e_i \rangle$ by adding noise to the variables μ_j for $j \in [1, m+1]$. The differentially private approximation procedure based on the Analytic Gaussian mechanism is presented in Algorithm 1. Theorem 2 guarantees that Algorithm 1 satisfies (ϵ, δ) -DP. Moreover, this framework is compatible with a

variety of DP mechanisms and noise distributions beyond the Gaussian mechanism, providing flexibility for diverse scenarios.

Algorithm 1 Differentially Private Legendre Polynomial Projection

Input: eCDF F_n , Legendre polynomial basis $\{P_i\}_{i=0}^m$, and the ℓ_2 sensitivity $\Delta_2 = \sqrt{\frac{5m+8}{2n^2}}$ of μ .

Output: Privacy-preserving coefficients $\{\tilde{c}_i\}_{i=0}^m$ and $\tilde{F}_n$.

Do:

1. Compute the moment vector $\mu = [\mu_1, \mu_2, \dots, \mu_{m+1}]^\top$ where $\mu_j = \frac{1}{n} \sum_{k=1}^n x_k^j$ for $j \in [1, m+1]$.
2. Add noise $\tilde{\mu} = \mu + \mathbf{z}$, where $\mathbf{z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$. The noise scale σ^2 is computed based on the sensitivity Δ_2 and [33, Algorithm 1].
3. Calculate the DP coefficients $\tilde{c}_i = \alpha_i \sum_{j=0}^i \beta_{i,j} (1 - \tilde{\mu}_{j+1})$ for $i \in [0, m]$.
4. Construct the DP eCDF: $\tilde{F}_n(x) = \sum_{i=0}^m \tilde{c}_i e_i(x)$.

End

Theorem 2. *Algorithm 1 satisfies (ϵ, δ) -differential privacy.*

Proof. According to Lemma 6 in Appendix B, the ℓ_2 sensitivity of μ is $\Delta_2 = \sqrt{\frac{5m+8}{2n^2}}$. Adding Gaussian noise calibrated via the Analytic Gaussian mechanism [33] with scale based on Δ_2 ensures that Algorithm 1 satisfies (ϵ, δ) -differential privacy. $\square$

To evaluate the noisy approximation $\tilde{F}_n$, Theorem 3 provides an upper bound on its distance from the true CDF F . Specifically, the theorem shows that if the polynomial space $\mathcal{P}$ is a suitable approximation space, meaning $\hat{F} \in \mathcal{P}$ adequately represents the true CDF, then the noisy approximation of the eCDF will also remain close to the true CDF.

Theorem 3 (Upper Bound for $\|F - \tilde{F}_n\|_2$). *Let F be the true CDF of a random variable with $x \in [-1, 1]$. If $\hat{F}$ is the optimal approximation of F in the polynomial space $\mathcal{P}$ and $\|F - \hat{F}\|_2 \leq \alpha$, then with probability at least $1 - 2 \exp\left(-\frac{n(\tau-\alpha)^2}{16}\right) - 2(m+1) \exp\left(-\frac{(\tau-\alpha)^2}{4(m+1)^4 \sigma^2}\right)$, we have $\|F - \tilde{F}_n\|_2 \leq \tau$ for $\tau > \alpha > 0$.*

Proof. Let F be the true CDF of a random variable with support $x \in [-1, 1]$, then $F \in L^\infty[-1, 1]$ and $F \in L^2[-1, 1]$. Let $\hat{F}$ be the optimal approximation of F in the polynomial space $\mathcal{P}$ assuming $\|F - \hat{F}\|_2 \leq \alpha$ for some $\alpha > 0$. The upper bound for $\|F - \tilde{F}_n\|_2$ is then given by

$$\begin{aligned} \|F - \tilde{F}_n\|_2 &\leq \|F - F_n\|_2 + \|F_n - \hat{F}_n\|_2 + \|\hat{F}_n - \tilde{F}_n\|_2 \\ &\leq \|F - F_n\|_2 + \|F_n - \hat{F}\|_2 + \|\hat{F}_n - \tilde{F}_n\|_2 \\ &\leq \|F - \hat{F}\|_2 + 2\|F - F_n\|_2 + \|\hat{F}_n - \tilde{F}_n\|_2 \\ &\leq \alpha + 2\sqrt{2}\|F - F_n\|_\infty + \sqrt{2}\|\hat{F}_n - \tilde{F}_n\|_\infty. \end{aligned} \quad (1)$$

The first and third inequalities follow from the triangle inequality, the second inequality follows from Theorem 7, and the last inequality uses the bound $\|f\|_2 \leq \sqrt{2}\|f\|_\infty$. By leveraging the triangle inequality and the fact that $\hat{F}$ is the optimal approximation of the true distribution, we avoid

directly computing the difference between $\hat{F}$ and $\tilde{F}_n$. The bound for $\|\hat{F}_n - \tilde{F}_n\|_\infty$ is (see Lemma 8 in Appendix B)

$$\|\hat{F}_n - \tilde{F}_n\|_\infty \leq \frac{(m+1)^2}{2} \max_{i \in [1, m+1]} |z_i|. \quad (2)$$

By combining (1) and (2), we obtain the following result:

$$\begin{aligned} & \mathbb{P}(\|F - \tilde{F}_n\|_2 > \tau) \\ & \leq \mathbb{P}(\alpha + 2\sqrt{2}\|F - F_n\|_\infty + \sqrt{2}\|\hat{F}_n - \tilde{F}_n\|_\infty > \tau) \\ & \leq \mathbb{P}\left(\|F - F_n\|_\infty > \frac{\tau - \alpha}{4\sqrt{2}}\right) + \mathbb{P}\left(\|\hat{F}_n - \tilde{F}_n\|_\infty > \frac{\tau - \alpha}{2\sqrt{2}}\right) \\ & \leq 2 \exp\left(-\frac{n(\tau - \alpha)^2}{16}\right) + 2(m+1) \exp\left(-\frac{(\tau - \alpha)^2}{4(m+1)^4\sigma^2}\right). \end{aligned} \quad (3)$$

The inequality in the last step is due to the DKW inequality [44] and an upper bound for the the maxima of subgaussian random variables [50]. $\square$

Remark 1. The magnitude of the approximation term α and its decay with the polynomial degree m depend on the regularity of the underlying CDF. Specifically, approximation results for orthogonal polynomial projections show that, if F belongs to the Sobolev space $H^r([-1, 1])$ for some $r > 0$, then $\alpha \leq C_r(m+1)^{-r}\|F\|_{H^r([-1, 1])}$, where $C_r > 0$ is independent of m [51], [52]. In particular, $F \in H^1([-1, 1])$ whenever the distribution admits a square-integrable density on $[-1, 1]$. More generally, if the density $f = F'$ has higher Sobolev regularity, then the CDF F inherits correspondingly higher regularity, leading to a faster decay of the approximation error with m . Thus, smoother distributions can be approximated accurately with lower-degree polynomial spaces.

Remark 2. Equation (3) indicates that the difference between the DP eCDF and the true CDF is governed by two bounds. The first term in (3) represents the distance between the true CDF and the eCDF, while the second term represents the distance between the approximation of the eCDF and its noisy counterpart. As the number of data points n increases, the first term decreases due to the convergence of the eCDF to the true CDF. The second term also decreases because the noise required diminishes as the sensitivity Δ_2 decreases. A larger number of polynomials m does not necessarily lead to a better result. While a higher m may improve the approximation of the eCDF, it also increases the noise due to a higher sensitivity.

The proposed polynomial projection framework can also be extended to high-dimensional CDF estimation using tensor-product polynomial spaces. Theorem 6 in Appendix B provides a closed-form expression for the corresponding projection coefficients in terms of multivariate empirical moments, while Lemma 7 in Appendix B characterizes their sensitivity for privacy-preserving estimation.

C. Effect of Post-processing

Similar to the HQ and AQ methods discussed earlier, our approach also requires post-processing, since the obtained $\tilde{F}_n$ is not guaranteed to be monotonically non-decreasing on

$[0, 1]$. We adopt isotonic regression [53] as the post-processing method. Proposition 1 (see Appendix B for proof) shows that this method brings the noisy approximation closer to the true CDF without compromising the accuracy of distribution estimation. Although isotonic regression is traditionally formulated under the L^2 norm, it can be extended to other L^p norms.

Proposition 1. Let $\tilde{F}_n \in L^2([-1, 1])$, and let $F : [-1, 1] \rightarrow [0, 1]$ be a monotone non-decreasing function. Define

$$\tilde{F}_n^{\text{iso}} = \arg \min_{f \in \mathcal{C}} \|f - \tilde{F}_n\|_2^2,$$

where the constraint set is $\mathcal{C} = \mathcal{C}_1 \cap \mathcal{C}_2$, with

$$\begin{aligned} \mathcal{C}_1 &= \{f \in L^2([-1, 1]) \mid f(x_1) \leq f(x_2) \text{ for a.e. } x_1 \leq x_2\}, \\ \mathcal{C}_2 &= \{f \in L^2([-1, 1]) \mid f(x) \in [0, 1] \text{ for a.e. } x \in [-1, 1]\}. \end{aligned}$$

Then the following inequality holds:

$$\|\tilde{F}_n^{\text{iso}} - F\|_2 \leq \|\tilde{F}_n - F\|_2.$$

At first glance, all this result says is that postprocessing using isotonic regression does not hurt the accuracy. Empirically, however, it can make a significant difference.

V. SPARSE APPROXIMATION VIA MATCHING PURSUIT

The DP polynomial projection method in the previous question is essentially an “off-the-shelf” approach which does not rely on the properties of real CDF functions, such as monotonicity and the values at endpoints. Increasing the number of basis functions does not necessarily improve the approximation, since it also amplifies the noise required for privacy. On the other hand, using too few basis functions may lead to a poor approximation of the true CDF. To balance this trade-off, we consider a function space $\mathcal{G}$ spanned by a large dictionary of m arbitrary functions, which need not be orthonormal. We then select the top s functions with the largest absolute inner products with the empirical distribution. Our theoretical analysis focuses on the case of orthonormal dictionaries. The experiments consider both orthonormal dictionaries, which align with the theoretical setting, and non-orthogonal dictionaries. The latter are investigated as empirical extensions to explore the effects of different dictionary constructions and potential extensions of the framework beyond the current theoretical setting. The empirical performance of different dictionary constructions is discussed in Section VI-C. Lemma 1 (see Appendix B for proof) shows that choosing s functions from a larger pool of m orthonormal basis functions achieves a smaller approximation error compared to using a fixed set of s basis functions. Although a larger dictionary increases computational cost, it enhances the expressive power of the representation.

Lemma 1. Let $\mathcal{D}_m = \{\phi_i\}_{i=1}^m$ be an orthonormal basis for a subspace of $L^2([-1, 1])$. For any target function $f \in L^2([-1, 1])$, define:

- (1) Fixed basis approximation: $\hat{f} = \sum_{i=1}^s \langle f, \phi_i \rangle \phi_i$, where $\{\phi_i\}_{i=1}^s$ are the first s basis elements in $\mathcal{D}_m$.

(2) *Adaptive selection*: $\hat{f}^s = \sum_{i=1}^s \langle f, \phi_{I_i} \rangle \phi_{I_i}$, where $\{\phi_{I_i}\}_{i=1}^s$ corresponds to the s basis functions with the largest absolute inner products $|\langle f, \phi_i \rangle|$.

Then the following inequality holds:

$$\|f - \hat{f}^s\|_2 \leq \|f - \hat{f}\|_2,$$

with strict inequality unless $\{\phi_{I_i}\}_{i=1}^s = \{\phi_i\}_{i=1}^s$.

To implement this selection process, we adopt a sparse representation approach, namely matching pursuit (MP) [54], a classical algorithm widely used in dictionary learning. Formally, given an orthonormal dictionary $\mathcal{D} = \{\phi_i\}_{i=1}^m \subset L^2([-1, 1])$, the MP algorithm iteratively selects the indices $\{I_i\}_{i=1}^s$ corresponding to the s most relevant basis functions. Relevance is quantified by the inner product between each candidate basis function and the current residual, defined as the unexplained portion of the target function after subtracting the contributions of the previously selected basis functions. This yields the sparse approximation

$$\hat{F}_n^s(x) = \sum_{i=1}^s c_i \phi_{I_i}(x),$$

where ϕ_{I_i} are dynamically selected from $\mathcal{D}$ and c_i denote the inner products with the residuals at each iteration. Lemma 2 (see Appendix B for proof) establishes that, when the dictionary is orthonormal, this procedure is equivalent to selecting the s basis functions with the largest coefficients from the full dictionary. Moreover, our approach naturally extends to dictionaries with non-orthonormal atoms, in which case the MP procedure remains applicable.

Lemma 2. *If $\mathcal{D} = \{\phi_i\}_{i=1}^m$ is an orthonormal basis, then for any target function f , MP algorithm with sparsity level s selects the s basis functions with the largest absolute inner products $|\langle f, \phi_i \rangle|$. In this case, the result of MP is equivalent to selecting the top s basis elements ranked by $|\langle f, \phi_i \rangle|$.*

To ensure differential privacy in the selection of I_i and its coefficient c_i , we employ the report noisy max (RNM) mechanism [55]. The resulting differentially private approximation (see Algorithm 2) is

$$\tilde{F}_n^s(x) = \sum_{i=1}^s \tilde{c}_i \phi_{\tilde{I}_i}(x),$$

where $\tilde{I}_i$ and $\tilde{c}_i$ denote the privacy-preserving index and coefficient, respectively. Lemma 3 and Lemma 4 (see Appendix B for proof) analyze the sensitivity of the (absolute) coefficients, which is crucial for determining the appropriate noise level required to ensure differential privacy. Based on these results, Theorem 4 establishes that Algorithm 2 satisfies $(2s\epsilon', 0)$ -DP under the basic composition rule. Although more advanced composition methods could be applied to relax the privacy cost by introducing a non-zero δ . Finally, the utility of the proposed method is analyzed in Theorem 5, which provides a bound on the estimation error introduced by the added noise.

Lemma 3 (Sensitivity of Inner Product). *Let $\langle \tilde{r}_i, \phi \rangle$ denote the inner product between the residual $\tilde{r}_i$ at the i -th step and*

Algorithm 2 Differentially private approximation via matching pursuit

Input: eCDF F_n , dictionary $\mathcal{D} = \{\phi_i\}_{i=1}^m$, sparsity level s , privacy parameter ϵ' and sensitivity Δ_{MP} .

Output: List of privacy-preserving coefficients $\{\tilde{c}_i\}_{i=1}^s$ and indices for corresponding atoms $\{\tilde{I}_i\}_{i=1}^s$.

Initialization: $\tilde{r}_1 \leftarrow F_n$.

For $i = 1$ **to** s :

1. Find $\phi_{\tilde{I}_i} \in \mathcal{D}$ using the RNM mechanism with Laplace noise of scale $b = \Delta_{MP}/\epsilon'$:

$$\phi_{\tilde{I}_i} = \arg \max_{\phi \in \mathcal{D}} (|\langle \tilde{r}_i, \phi \rangle| + \text{Laplace}(0, b)).$$

2. Compute the noisy coefficient: $\tilde{c}_i \leftarrow \langle \tilde{r}_i, \phi_{\tilde{I}_i} \rangle + \text{Laplace}(0, b)$.

3. Update the residual: $\tilde{r}_{i+1} \leftarrow \tilde{r}_i - \tilde{c}_i \phi_{\tilde{I}_i}$.

4. $i \leftarrow i + 1$.

End

an arbitrary basis function ϕ . Assume the residual is defined as

$$\tilde{r}_i = F_D - \sum_{j=1}^{i-1} \tilde{c}_j \phi_{\tilde{I}_j},$$

where F_D is the eCDF of the dataset $D = \{x_k\}_{k=1}^n$, and $\tilde{c}_j$ are the coefficients computed in previous steps. Then, the sensitivity of this inner product is

$$\Delta_{MP} = \frac{1}{n} \int |\phi(x)| dx.$$

Lemma 4 (Sensitivity of Absolute Inner Product). *Let $|\langle \tilde{r}_i, \phi \rangle|$ denote the absolute inner product between the residual $\tilde{r}_i$ at the i -th step and an arbitrary basis function ϕ . Then, the sensitivity of the absolute inner product is*

$$\Delta_{AMP} = \Delta_{MP}.$$

Theorem 4. *Let ϵ' denote the privacy budget allocated to each step, and let $\epsilon = 2s\epsilon'$. Then, Algorithm 2 satisfies ϵ -DP, where s denotes the sparsity level.*

Proof. Each iteration of Algorithm 2 involves two operations that require differential privacy: selecting the basis index and releasing the selected coefficient. Each of these operations uses ϵ' -DP. Over s iterations, the total privacy cost is $2s\epsilon'$ -DP, according to the basic composition theorem. $\square$

Theorem 5 (Upper Bound for $\|F - \tilde{F}_n^s\|_2$). *Fix a privacy parameter $\epsilon > 0$, sparsity level s , and sample size n . Let $\mathcal{D} = \{\phi_i\}_{i=1}^m$ denote a family of orthonormal functions on $[-1, 1]$, and let $\mathcal{G}$ be the linear span of $\mathcal{D}$. Let F be the true CDF, and let F_n denote the eCDF formed from n i.i.d. samples from F . Let $\tilde{F}^s$ and $\tilde{F}_n^s$ denote the optimal non-private s -*

sparse approximations of the true CDF F and the empirical CDF F_n in $\mathcal{G}$, respectively, where $\hat{F}_n^s$ has the form

$$\hat{F}_n^s(x) = \sum_{i=1}^s c_i \phi_{I_i}(x),$$

and the index set $\{I_1, \dots, I_s\}$ corresponds to the s functions with the largest absolute inner products with F_n . Let $\tilde{F}_n^s$ be the output of Algorithm 2 with input F_n , s , ϵ , and Δ_{MP} . Finally, define $\beta = \sqrt{2} \sum_{i=1}^s |c_i|$. Then if $\|F - \hat{F}_n^s\|_2 \leq \alpha$, with probability at least $1 - 2 \exp\left(-\frac{n(\tau - \alpha - \beta)^2}{16}\right) - \exp\left(s - \frac{n\epsilon(\tau - \alpha - \beta)}{2\|\phi\|_1}\right) \left(\frac{n\epsilon(\tau - \alpha - \beta)}{2s\|\phi\|_1}\right)^s$, we have $\|F - \tilde{F}_n^s\|_2 \leq \tau$ for $\tau > \alpha + \beta > 0$ and $\tau - \alpha - \beta > 2s\|\phi\|_1/n\epsilon$.

Proof. Let $\tilde{F}_n^s = \sum_{i=1}^s c_i \phi_{\tilde{I}_i}(x)$, where $\tilde{I}_i$ denotes the index selected via RNM. The total error between the true CDF and its privacy-preserving counterpart can be decomposed as follows:

$$\begin{aligned} \|F - \tilde{F}_n^s\|_2 &\leq \|F - F_n\|_2 + \|F_n - \hat{F}_n^s\|_2 + \|\hat{F}_n^s - \tilde{F}_n^s\|_2 \\ &\leq \|F - F_n\|_2 + \|F_n - \hat{F}_n^s\|_2 + \|\hat{F}_n^s - \tilde{F}_n^s\|_2 \\ &\leq \underbrace{\|F - \hat{F}_n^s\|_2}_{\text{approximation error}} + 2 \underbrace{\|F - F_n\|_2}_{\text{empirical error}} \\ &\quad + \underbrace{\|\hat{F}_n^s - \tilde{F}_n^s\|_2}_{\text{index error}} + \underbrace{\|\tilde{F}_n^s - \tilde{F}_n^s\|_2}_{\text{coefficient error}}. \end{aligned}$$

In the context of the MP algorithm, privacy error arises from two sources: the index error and the coefficient error. We now derive worst-case upper bounds for both components. Specifically, we assume that index perturbation occurs at every step, i.e., $I_i \neq \tilde{I}_i$ for all i , and we do not assume orthogonality among the perturbed basis functions $\tilde{I}_i$. In this case, we derive the following bounds:

$$\begin{aligned} \|\hat{F}_n^s - \tilde{F}_n^s\|_2 &= \left\| \sum_{i=1}^s c_i \phi_{I_i} - \sum_{i=1}^s c_i \phi_{\tilde{I}_i} \right\|_2 \\ &\leq \sum_{i=1}^s |c_i| \|\phi_{I_i} - \phi_{\tilde{I}_i}\|_2 \\ &= \sqrt{2} \sum_{i=1}^s |c_i| \cdot \mathbb{I}(I_i \neq \tilde{I}_i) \\ &\leq \sqrt{2} \sum_{i=1}^s |c_i|. \end{aligned}$$

Let $\beta = \sqrt{2} \sum_{i=1}^s |c_i|$ denote an upper bound for the index error. We now turn to the coefficient error term, which satisfies:

$$\begin{aligned} \|\tilde{F}_n^s - \tilde{F}_n^s\|_2 &= \left\| \sum_{i=1}^s c_i \phi_{\tilde{I}_i} - \sum_{i=1}^s \tilde{c}_i \phi_{\tilde{I}_i} \right\|_2 \\ &\leq \sum_{i=1}^s |c_i - \tilde{c}_i| \|\phi_{\tilde{I}_i}\|_2 \\ &= \sum_{i=1}^s |z_i|, \end{aligned}$$

where $z_i \sim \text{Laplace}(0, \Delta_{MP}/\epsilon)$ and $\Delta_{MP} = \frac{1}{n} \int |\phi(x)| dx = \frac{1}{n} \|\phi\|_1$. Then for any $\tau > \alpha + \beta$ and $\frac{\tau - \alpha - \beta}{2} > \frac{s\Delta_{MP}}{\epsilon}$, we obtain

$$\begin{aligned} &\mathbb{P}(\|F - \tilde{F}_n^s\|_2 > \tau) \\ &\leq \mathbb{P}(\alpha + 2\sqrt{2}\|F - F_n\|_\infty + \beta + \|\tilde{F}_n^s - \tilde{F}_n^s\|_2 > \tau) \\ &\leq \mathbb{P}\left(\|F - F_n\|_\infty > \frac{\tau - \alpha - \beta}{4\sqrt{2}}\right) \\ &\quad + \mathbb{P}\left(\sum_{i=1}^s |z_i| > \frac{\tau - \alpha - \beta}{2}\right) \\ &\leq 2 \exp\left(-\frac{n(\tau - \alpha - \beta)^2}{16}\right) \\ &\quad + \exp\left(s - \frac{n\epsilon(\tau - \alpha - \beta)}{2\|\phi\|_1}\right) \left(\frac{n\epsilon(\tau - \alpha - \beta)}{2s\|\phi\|_1}\right)^s. \end{aligned}$$

The first term in the last inequality comes from the DKW inequality. The second term arises because $z_i \sim \text{Laplace}(0, \Delta_{MP}/\epsilon)$, which implies that $|z_i| \sim \text{Exponential}(\epsilon/\Delta_{MP})$. Consequently, the sum follows an Erlang distribution, i.e., $\sum_{i=1}^s |z_i| \sim \text{Erlang}(s, \epsilon/\Delta_{MP})$, where ϵ/Δ_{MP} is the rate parameter. Based on Lemma 5 in Appendix B, we obtain the second term in the bound. $\square$

Remark 3. Theorem 5 shows that increasing the sparsity level s , and thus using more functions to approximate the eCDF, does not necessarily reduce the distance between the true CDF and its noisy approximation, as a larger s increases the likelihood of incorrect index and coefficient selection. Similarly, enlarging the dictionary size m does not necessarily bring the DP CDF closer to the true CDF, since a richer dictionary reduces the approximation error α but simultaneously increases the noise-induced error β . By contrast, increasing the sample size n consistently decreases the distance between the CDF and its approximation, because the empirical error diminishes and the magnitude of the added noise is reduced.

VI. EXPERIMENTS

In this experimental section², we investigate the impact of key parameters, including sparsity level and dictionary size, on CDF approximation (see Section VI-A). We then provide a comparative analysis of different methods (see Section VI-B) and finally discuss the effect of varying dictionary compositions (see Section VI-C).

A. Effect of Parameters

To quantitatively study the effect of parameters, we assess the distance between the CDF and its (noisy) approximation using three measures: the Kolmogorov-Smirnov distance [56], the earth mover's distance [57], and the energy distance [58]. The Kolmogorov-Smirnov distance is defined as the maximum difference between the two functions. In contrast, the earth mover's distance, given in the univariate case by $W(f_1, f_2) = \int |f_1 - f_2|$, measures the integral of the absolute difference between the CDFs. The energy distance, expressed as $E(f_1, f_2) = \sqrt{2 \int (f_1 - f_2)^2}$, penalizes larger deviations between the distributions more heavily.

²The implementation code is available at <https://github.com/yetaoyl/dp-cdf>.

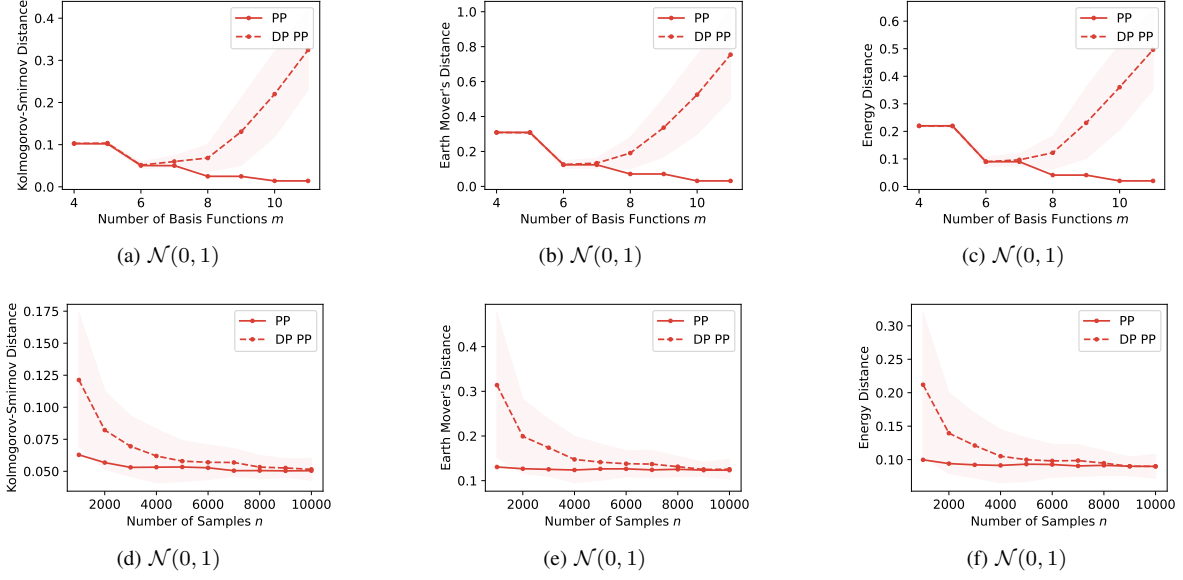


Fig. 2: Comparison of distances between PP-based approximation and the true CDF under two experimental settings. (a)-(c) Effect of the number of basis functions m with $n = 10^4$; (d)-(f) Effect of the sample size n with $m = 6$. Experiments were repeated 50 times with $\epsilon = 0.5$ under pure differential privacy.

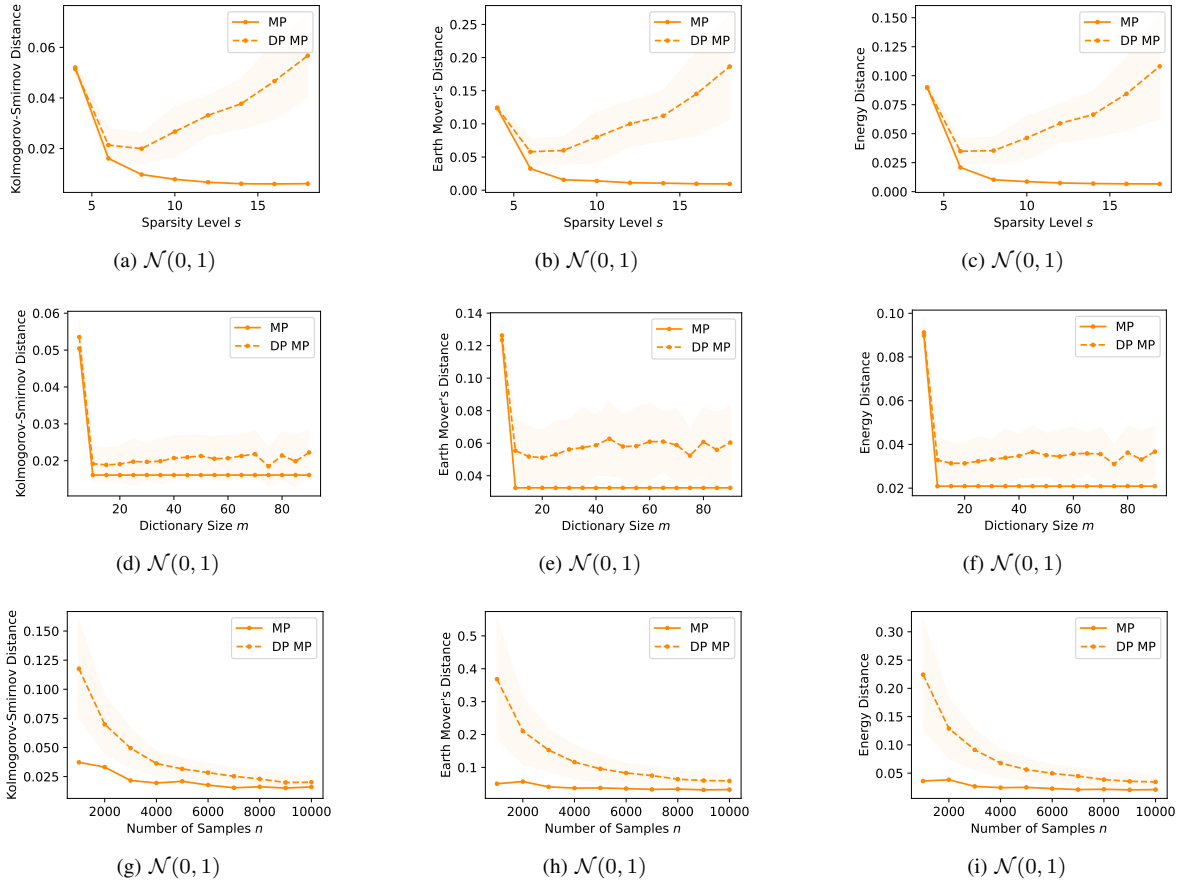


Fig. 3: Comparison of distances between MP-based approximation and the true CDF under three experimental settings. (a)-(c) Effect of the sparsity level s with $n = 10^4$ and a dictionary of $m = 40$ Legendre polynomials; (d)-(f) Effect of the dictionary size m with $n = 10^4$ and $s = 6$; (g)-(i) Effect of the sample size n with a dictionary of $m = 40$ Legendre polynomials and $s = 6$. Experiments were repeated 50 times with $\epsilon = 0.5$ under pure differential privacy.

1) *Polynomial Projection*: Figure 2 (a)-(c) and Figure 12 (a)-(f) in Appendix C illustrate the effect of m on approximation performance. In the non-private setting, increasing m consistently reduces the error between the projected eCDF and the true CDF, as the polynomial space becomes more expressive. In the differentially private setting, however, the error first decreases and then increases. This is because a richer polynomial space initially improves approximation quality, but a larger m requires releasing more coefficients, which in turn necessitates adding more noise. Consequently, the discrepancy between the approximated and true CDF tends to grow when m becomes large. Based on experimental results, values of m between 5 and 8 are empirically preferable.

The effect of the sample size n on approximation performance is shown in Figure 2 (d)-(f) and Figure 12 (g)-(l) in Appendix C. In the non-private setting, the error remains nearly constant as n increases. This reflects the inherent approximation limit imposed by the predefined polynomial space, since even the best projection within this space cannot perfectly match the true CDF. As mentioned previously, increasing n reduces the error in differentially private settings, and the performance gradually approaches that of the non-private setting.

2) *Matching Pursuit*: The effect of the sparsity level s on the approximation performance of the matching pursuit method is illustrated in Figure 3 (a)-(c) and Figure 13 (a)-(f) in Appendix C. In the non-private setting, the error decreases as s increases and eventually converges to the approximation limit determined by the expressive power of the dictionary. In the differentially private setting, however, increasing s does not necessarily reduce the error. Similar to the PP method, using more basis functions requires injecting additional noise into the released indices and coefficients, which may offset the potential gains in approximation accuracy. We also study the effect of the dictionary size m , as illustrated in Figure 3 (d)-(f) and Figure 13 (g)-(l) in Appendix C, and find that enlarging m does not necessarily reduce the error in the differentially private setting. Consistent with the PP method, increasing the sample size n consistently reduces the approximation error in the private setting, as shown in Figure 3 (g)-(i) and Figure 13 (m)-(r) in Appendix C.

B. Comparison of Methods

To evaluate the performance of our method, we conducted experiments on both synthetic and real-world datasets. Figure 4 and Figure 14 in Appendix C present comparative examples of our methods alongside existing baselines. The first column shows the HQ method, the second column shows the TB method, the third column shows the AQ method, and the last two columns depict our proposed methods: polynomial projection and sparse approximation via matching pursuit. With a small bin number in the HQ method, noticeable discrepancies are observed between the CDF approximation (green solid line) and the true CDF (gray solid line), even without perturbation. The TB method provides a more accurate approximation by using a finer hierarchical partition. The AQ method yields an accurate CDF approximation (blue solid

line), but its privacy-preserving counterpart (blue dashed line) tends to focus on regions with the steepest slope, as these regions correspond to greater changes in the quantiles. Our methods accurately fit the true CDF, regardless of whether privacy protection is considered. Notably, the matching pursuit approach, which leverages a larger dictionary and sparse function selection, achieves a more accurate approximation than the polynomial projection method, as indicated by the red and orange solid lines.

To quantitatively compare the performance of different methods, Figure 5 and Figure 15 in Appendix C present experimental results across various distributions and datasets. For ease of comparison, all experiments are conducted under pure differential privacy with $\delta = 0$. In practice, more advanced composition techniques can be applied with a small nonzero δ to achieve tighter privacy accounting and improve utility. We systematically investigate the hyperparameters of each method using grid search and select representative values for subsequent experiments based on the resulting error between the private CDF and the true CDF. For example, for HQ, we vary the number of bins from 10 to 100 in increments of 10. The relative performance of the baseline methods varies across distributions and evaluation metrics. The HQ method generally exhibits relatively large errors, while the TB method can perform particularly well. For smooth unimodal distributions, AQ struggles to capture fine-grained variations, whereas for distributions with pronounced local variations, it achieves better approximation. Overall, our proposed PP and MP methods demonstrate competitive and consistent performance across varying privacy levels.

Algorithm 3 Decentralized Differentially Private Legendre Polynomial Projection

Input: Data held by S sites, Legendre polynomial basis $\{P_i\}_{i=0}^m$, and privacy parameters ϵ and δ .

Output: Privacy-preserving coefficients $\{\tilde{c}_i\}_{i=0}^m$ and $\tilde{F}_n$.

Do:

1. For each site $q \in [1, S]$, let n_q be the site sample size and compute the local moment vector $\boldsymbol{\mu}^{(q)} = [\mu_1^{(q)}, \mu_2^{(q)}, \dots, \mu_{m+1}^{(q)}]^\top$, where $\mu_j^{(q)} = \frac{1}{n_q} \sum x_k^j$ for $j \in [1, m+1]$.
2. At each site, add noise to the local moment vector: $\tilde{\boldsymbol{\mu}}^{(q)} = \boldsymbol{\mu}^{(q)} + \mathbf{z}^{(q)}$, where the distribution and scale of $\mathbf{z}^{(q)}$ are determined by the selected privacy parameters and mechanism.
3. Each site sends $\tilde{\boldsymbol{\mu}}^{(q)}$ to the central server.
4. The central server computes the weighted aggregate $\tilde{\boldsymbol{\mu}} = \sum_{q=1}^S \frac{n_q}{\sum_{r=1}^S n_r} \tilde{\boldsymbol{\mu}}^{(q)}$.
5. Calculate the DP coefficients $\tilde{c}_i = \alpha_i \sum_{j=0}^i \beta_{i,j} (1 - \tilde{\mu}_{j+1})$ for $i \in [0, m]$.
6. Construct the DP eCDF: $\tilde{F}_n(x) = \sum_{i=0}^m \tilde{c}_i e_i(x)$.

End

Our methods offer additional benefits in decentralized settings. Suppose S sites collect sensitive data and send privatized information to an untrusted central server for CDF computation. When the AQ or MP method is applied, multiple

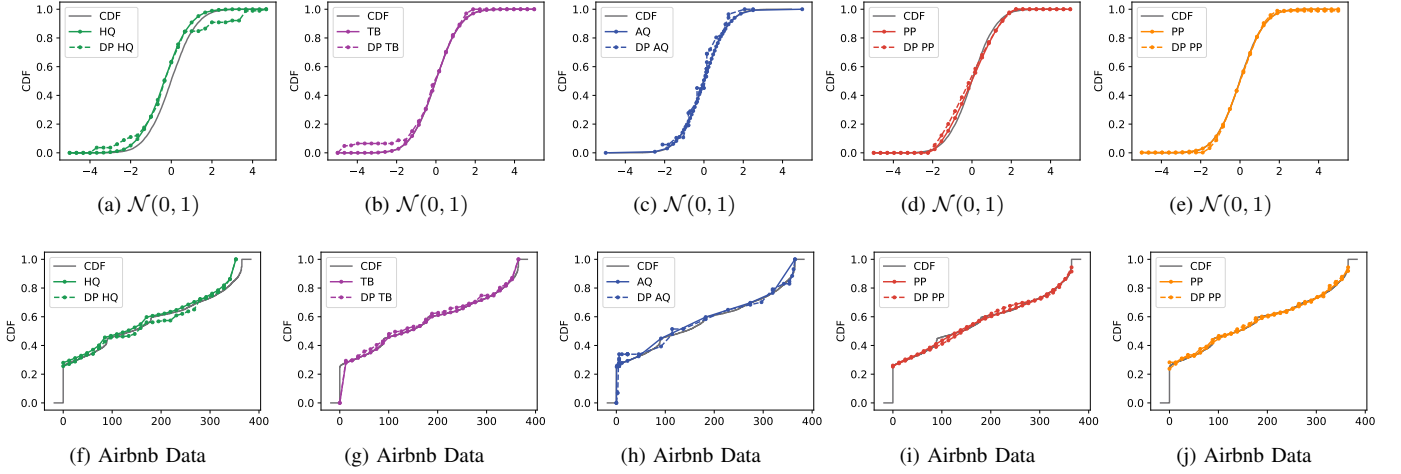


Fig. 4: Comparison of DP approximation methods for the normal distribution and the U.S. Airbnb data under pure differential privacy with $\epsilon = 0.1$. (a)-(e) Normal distribution $\mathcal{N}(0, 1)$ with $n = 10^4$; (f)-(j) U.S. Airbnb Data, where `availability_365` records the number of days within a year that a listing is available for booking. The original dataset contains 226030 records, from which a 10% subsample ($n = 22603$) is drawn. For both datasets, HQ uses 30 bins, TB uses 1024 leaves, AQ runs for 40 iterations, PP employs 6 basis functions, and MP selects 7 basis functions from a dictionary of 40 Legendre atoms.

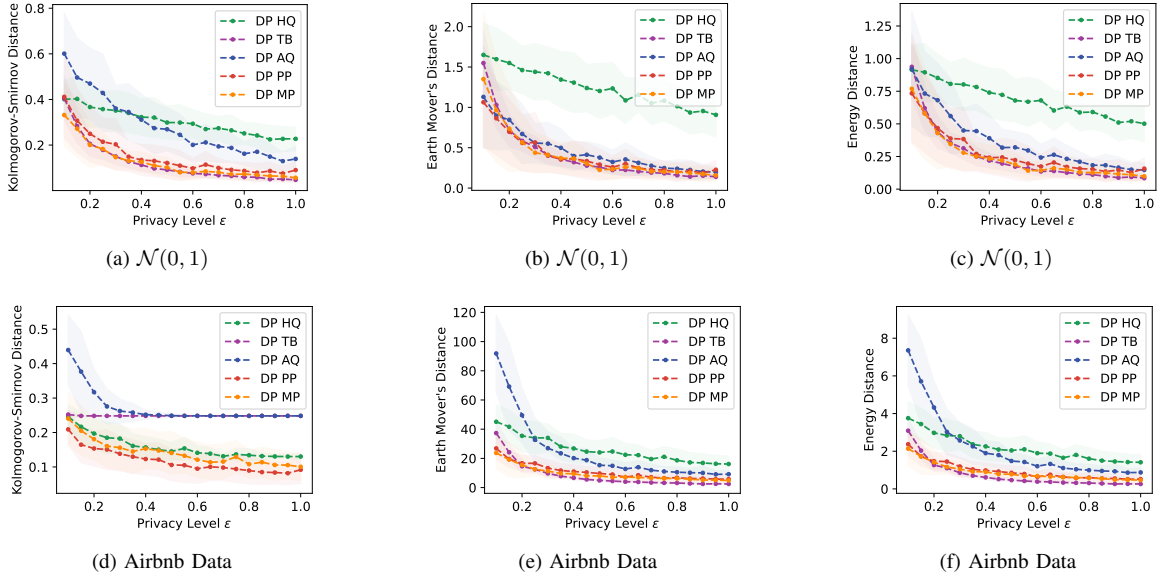


Fig. 5: Comparison of distances between different DP CDF methods and the true CDF under three metrics. (a)-(c) Normal distribution $\mathcal{N}(0, 1)$ with $n = 10^3$; (d)-(f) U.S. Airbnb Data with $n = 2260$. Each experiment was repeated 50 times under pure differential privacy, where ϵ denotes the overall privacy budget. For both datasets, HQ uses 40 bins, TB uses 1024 leaves, AQ runs for 40 iterations, PP employs 6 basis functions, and MP selects 5 basis functions from a dictionary of 40 Legendre atoms.

rounds of communication between the central server and each site are required. Under the same communication constraints, the MP method achieves a better approximation than AQ. In contrast, the HQ, TB, and PP methods require only a single round of communication, in which each site sends its noisy histogram, noisy tree representation, or noisy values of μ_i , $i \in [1, m + 1]$, respectively, to the central server. The decentralized implementations of our PP and MP methods are summarized in Algorithm 3 and Algorithm 4, respectively.

Figure 6 and Figure 16 in Appendix C demonstrate the performance of different methods in the decentralized setting. Our PP and MP methods perform particularly well for smooth distributions and achieve competitive performance across the evaluated settings. Considering its competitive approximation performance and simple one-round communication procedure, the PP method is a strong candidate for decentralized settings.

Our methods also offer practical benefits when incorporating newly collected data. Specifically, for PP, we compute the

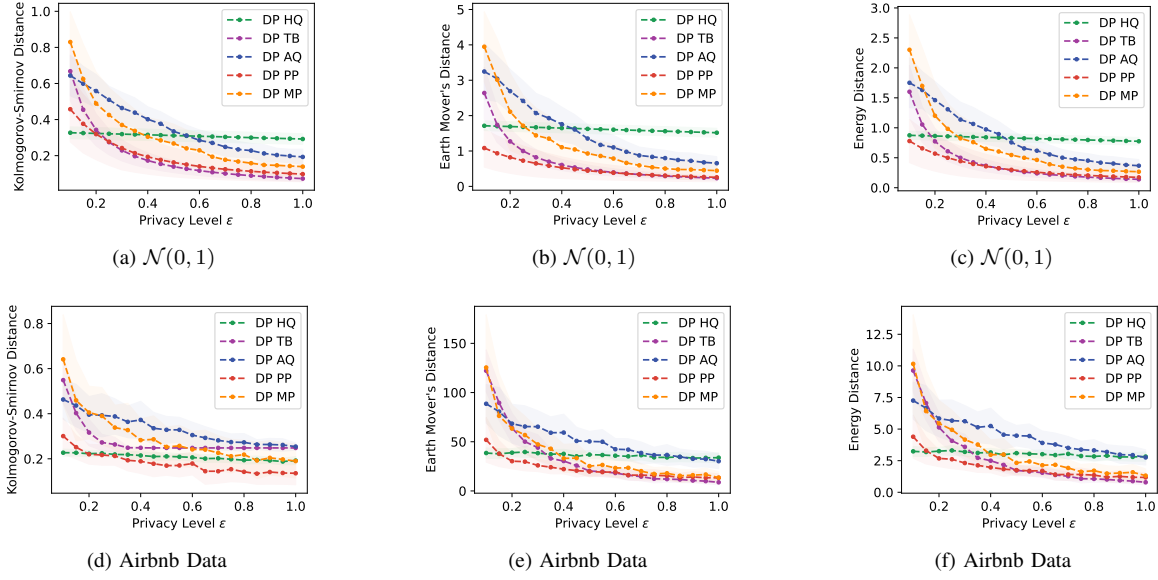


Fig. 6: Comparison of distances between different DP CDF methods and the true CDF under three metrics in a decentralized setting with 10 sites. (a)-(c) Normal distribution $\mathcal{N}(0, 1)$ with $n = 200$ samples per site; (d)-(f) U.S. Airbnb Data with $n = 226$ samples per site. Each experiment was repeated 50 times under pure differential privacy, where ϵ denotes the overall privacy budget. For both datasets, HQ uses 40 bins, TB uses 1024 leaves, AQ runs for 40 iterations, PP employs 6 basis functions, and MP selects 5 basis functions from a dictionary of 40 Legendre atoms.

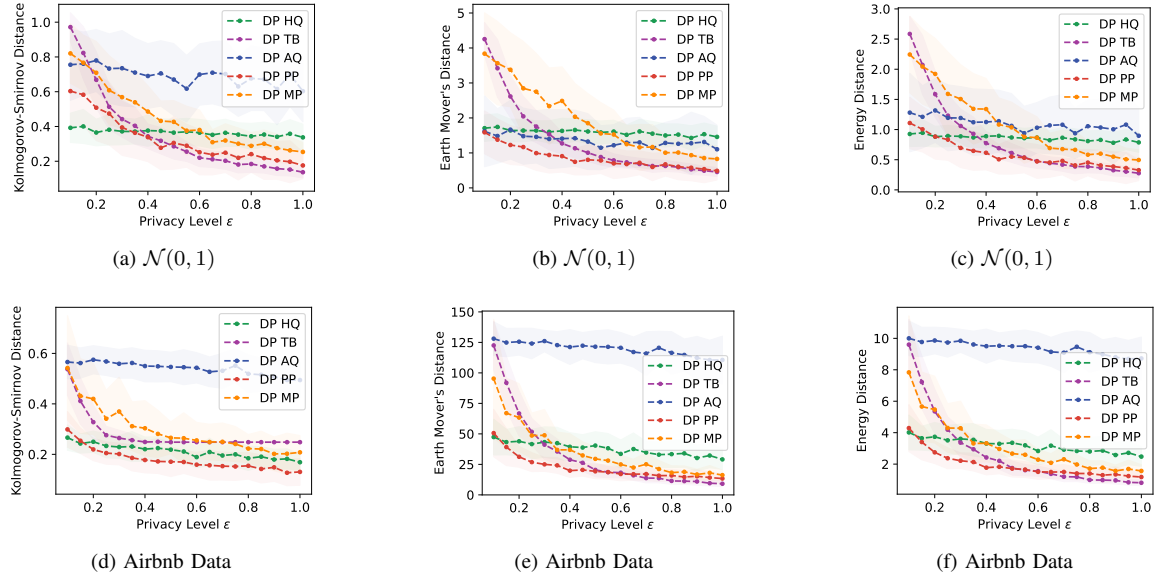


Fig. 7: Comparison of distances between different DP CDF methods and the true CDF under three metrics in a newly collected data setting, where the CDF is updated for a total of 10 rounds. (a)-(c) Normal distribution $\mathcal{N}(0, 1)$ with 100 newly collected samples per round; (d)-(f) U.S. Airbnb Data with 226 newly collected samples per round. Each experiment was repeated 50 times under pure differential privacy, where ϵ denotes the overall privacy budget. For both datasets, HQ uses 40 bins, TB uses 1024 leaves, AQ runs for 40 iterations, PP employs 6 basis functions, and MP selects 5 basis functions from a dictionary of 40 Legendre atoms.

noisy moment $\frac{1}{n_{\text{new}}} \sum_{k=1}^{n_{\text{new}}} x_k^j$ for the newly collected data and combine it with the previously stored noisy moment $\frac{1}{n_{\text{old}}} \sum_{k=1}^{n_{\text{old}}} x_k^j$ to obtain a noisy estimate of the updated moment $\frac{1}{n_{\text{old}} + n_{\text{new}}} \sum_{k=1}^{n_{\text{old}} + n_{\text{new}}} x_k^j$. Similarly, for MP, we compute the

residual between the CDF of the newly collected data and the previously stored CDF approximation, approximate this residual using the private matching pursuit procedure, and use the resulting approximation to update the stored CDF

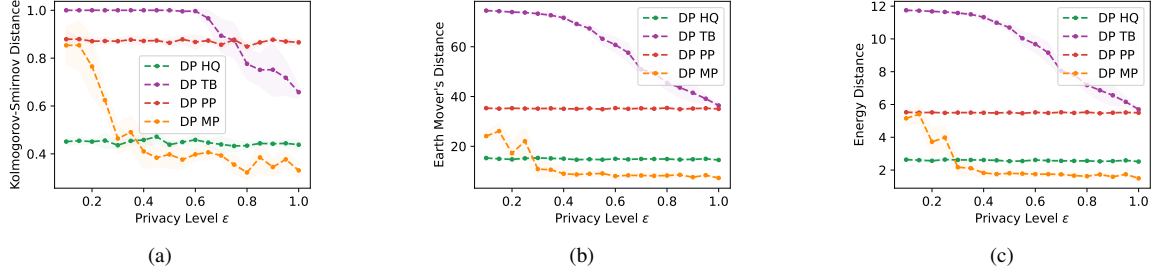


Fig. 8: Comparison of distances between different DP CDF methods and the true CDF under three metrics for a two-dimensional standard normal distribution with correlation coefficient $\rho = 0.5$. Each experiment was repeated 50 times with $n = 10^3$ under pure differential privacy, where ϵ denotes the overall privacy budget. HQ uses 40 bins per dimension, TB uses 512 leaves per dimension, PP employs 6 basis functions per dimension, and MP selects 5 basis functions from a dictionary of 80 atoms.

Algorithm 4 Decentralized Differentially Private Approximation via Matching Pursuit

Input: Data held by S sites, dictionary $\mathcal{D} = \{\phi_j\}_{j=1}^m$, sparsity level s , and privacy parameters.

Output: List of privacy-preserving coefficients $\{\tilde{c}_i\}_{i=1}^s$ and indices of the corresponding atoms $\{\tilde{I}_i\}_{i=1}^s$.

Initialization: $\tilde{F}_0 \leftarrow 0$, $\mathcal{I} \leftarrow \emptyset$.

For $i = 1$ **to** s :

1. At each site $q \in [1, S]$, compute the local residual $\tilde{r}_i^{(q)} = F_n^{(q)} - \tilde{F}_{i-1}$.
2. Each site privately selects an atom from the unselected dictionary: $\tilde{I}_i^{(q)} = \arg \max_{j \notin \mathcal{I}} (|\langle \tilde{r}_i^{(q)}, \phi_j \rangle| + z_{i,j}^{(q)})$, where the distribution and scale of $z_{i,j}^{(q)}$ are determined by the selected privacy parameters and mechanism.
3. Each site computes a noisy coefficient for the selected atom: $\tilde{c}_i^{(q)} = \langle \tilde{r}_i^{(q)}, \phi_{\tilde{I}_i^{(q)}} \rangle + \xi_i^{(q)}$.
4. Each site sends $\tilde{I}_i^{(q)}$ and $\tilde{c}_i^{(q)}$ to the central server.
5. The central server selects the atom receiving the largest number of votes: $\tilde{I}_i = \text{mode}(\tilde{I}_i^{(1)}, \dots, \tilde{I}_i^{(S)})$.
6. Let $\mathcal{S}_i = \{q : \tilde{I}_i^{(q)} = \tilde{I}_i\}$. The central server aggregates the noisy coefficients reported by the sites in $\mathcal{S}_i$: $\tilde{c}_i = \frac{1}{|\mathcal{S}_i|} \sum_{q \in \mathcal{S}_i} \tilde{c}_i^{(q)}$.
7. Update the global approximation: $\tilde{F}_i = \tilde{F}_{i-1} + \tilde{c}_i \phi_{\tilde{I}_i}$, and update $\mathcal{I} \leftarrow \mathcal{I} \cup \{\tilde{I}_i\}$.

End

approximation. Therefore, neither PP nor MP requires access to the previously collected raw data. In contrast, the AQ method requires revisiting the old data to update the CDF approximation, thereby increasing the privacy cost. The HQ and TB methods can also incorporate newly collected data without revisiting the old data as long as their underlying partitions remain fixed. Figure 7 and Figure 17 in Appendix C demonstrate the performance of different methods when incorporating newly collected data. The TB method performs well in this setting, while PP also achieves strong performance and generally outperforms MP. It should be noted that the strong performance of TB in these experiments assumes a fixed tree structure. If the tree structure changes, previously

collected data may need to be revisited, leading to additional privacy costs and potentially reduced utility. In addition, PP offers practical advantages. Reconstructing the DP CDF requires storing only a few values, determined by the number of basis functions m , whereas other methods typically need to store many more values depending on the bin count, tree size, iteration number, or the entire dictionary. Furthermore, PP directly provides moment information without extra computation.

While PP offers advantages in decentralized and newly collected data settings, MP is particularly well-suited to high-dimensional CDF approximation. As the dimensionality increases, the number of basis functions required by PP grows rapidly, making accurate approximation increasingly challenging. In contrast, MP selects only a small number of informative atoms from a larger dictionary, allowing it to exploit sparse representations without using all candidate basis functions. HQ and TB can also be extended to high-dimensional settings, however, the number of histogram bins or tree nodes grows rapidly with dimensionality, leading to increased estimation and privacy costs. Moreover, the AQ method considered in this work does not directly extend to multivariate CDF estimation due to the lack of a natural total ordering in multiple dimensions. Figure 8 shows that MP performs particularly well as the privacy budget increases.

C. Exploration of Dictionary Compositions

In this section, we explore three representative families of functions for constructing the dictionary: Legendre polynomials, B-splines, and normal distribution CDFs with varying means and variances. These choices illustrate distinct structural properties: Legendre polynomials are orthogonal, B-splines are non-orthogonal, and normal distribution CDFs are non-orthogonal and inherently monotonic. The dictionary is not limited to these three families and may be constructed from arbitrary functions, such as empirical CDFs derived from real-world data.

Figure 9 and Figure 18 in Appendix C present the reconstruction results. These results demonstrate that, as long as the dictionary is sufficiently rich, the CDF can be well approximated. Nevertheless, the approximation quality varies across

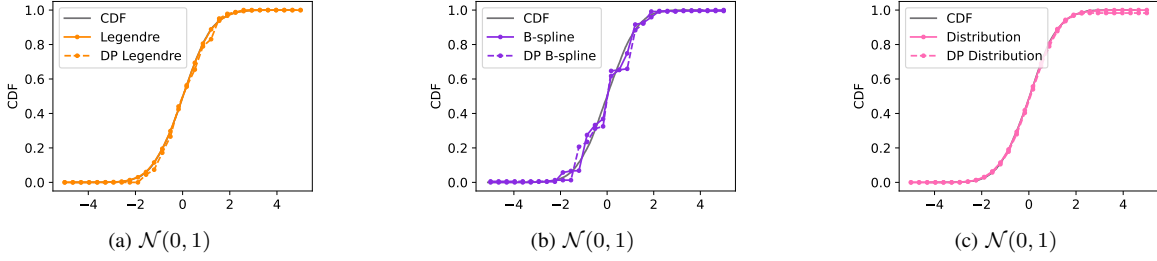


Fig. 9: Comparison of CDF reconstruction using different dictionaries with parameters $n = 10^4$, $\epsilon = 0.5$, $\delta = n^{-3/2}$, and sparsity level $s = 30$. (a) Dictionary constructed from 200 Legendre polynomials; (b) Dictionary constructed from 109 B-spline functions of degree 0 and 1; (c) Dictionary constructed from 400 normal CDFs with varying means and variances.

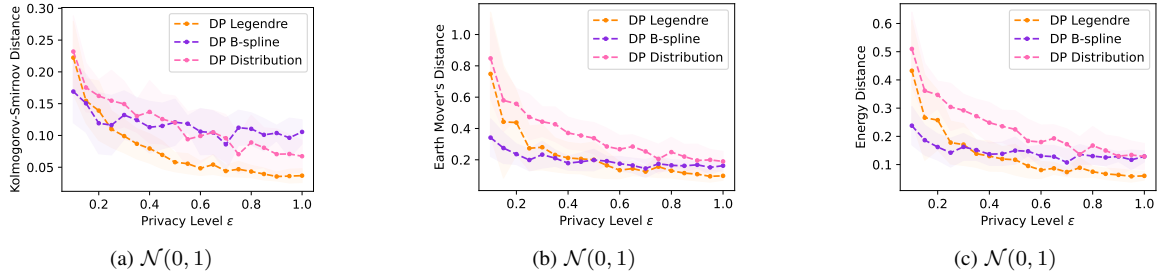


Fig. 10: Comparison of the approximation distances between the DP CDF obtained using different dictionaries and the true CDF. Each experiment was repeated 50 times with $n = 10^4$, $\delta = n^{-3/2}$, and sparsity level $s = 30$.

different dictionaries. B-splines perform particularly well in approximating complex multimodal distributions, as their local support enables flexible adaptation to local variations. In contrast, dictionaries constructed from normal distribution CDFs perform less effectively in such cases. This limitation arises because normal CDFs are inherently smooth and S-shaped, constrained to represent unimodal and symmetric distributions. Even when combined, they lack the flexibility to capture the step-like structures associated with multiple peaks. Moreover, their global support makes local adjustments inefficient, often requiring a large number of atoms to approximate complex patterns.

Figure 10 and Figure 19 in Appendix C show the approximation distance between the DP CDF and the true CDF. These results further confirm that, for complex distributions, dictionaries constructed from B-splines provide a more effective representation than the other two types.

VII. FUTURE DIRECTIONS

The main idea of this work lies in projecting the CDF into a predefined function space and applying differential privacy to the coefficients to protect the estimated CDF. Building on polynomial projection, we extend the framework to obtain sparse approximations from arbitrary function spaces constructed by dictionaries. Several promising directions remain open. Our current framework is primarily developed in the L^2 space, and extending the analysis to more general L^p spaces may provide broader insights. For MP with orthonormal dictionaries, the current error bound may be loose, motivating the development of a tighter bound. It is also important to

further investigate non-orthogonal dictionaries and develop corresponding theoretical guarantees. Although we have explored the extension to multivariate distributions, a more comprehensive analysis remains an important direction for future work. Another direction is to systematically investigate different post-processing techniques and their effects on the utility of the reconstructed CDF. It is also worthwhile to explore theoretical connections with robust statistics [59], whose modeling principles align with the foundations of differential privacy [60]–[62], potentially leading to universally valid privacy guarantees.

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